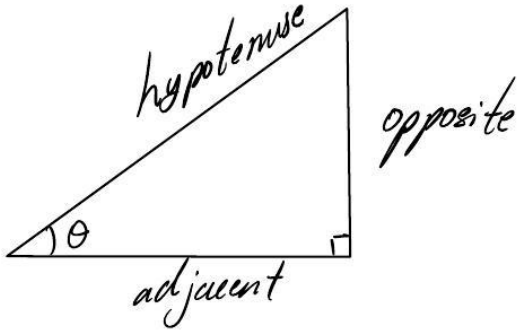
Lecture 8, Jan 8, 2025.

Appendix D (trigonometry).

The trigonometric functions.

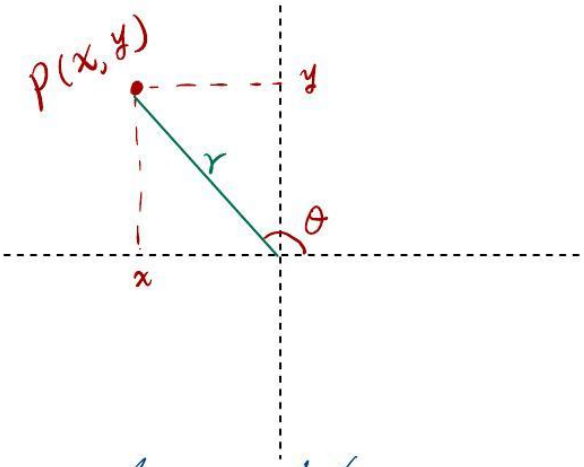


$$\sin\theta = \frac{opp}{hyp} \qquad \csc\theta = \frac{hyp}{opp}$$

$$\cos\theta = \frac{adj}{hyp} \qquad \sec\theta = \frac{hyp}{adj}$$

$$\tan\theta = \frac{opp}{adj} \qquad \cot\theta = \frac{adj}{opp}.$$

If $P = (x, y)$



$$\sin\theta = \frac{y}{r} \qquad \csc\theta = \frac{r}{y}$$

$$\cos\theta = \frac{x}{r} \qquad \sec\theta = \frac{r}{x}$$

$$\tan\theta = \frac{y}{x} \qquad \cot\theta = \frac{x}{y}.$$

trigonometric identities

i) $\csc\theta = \frac{1}{\sin\theta}$; $\sec\theta = \frac{1}{\cos\theta}$; $\cot\theta = \frac{1}{\tan\theta}$; $\tan\theta = \frac{\sin\theta}{\cos\theta}$; $\cot\theta = \frac{\cos\theta}{\sin\theta}$.

ii) $\sin^2\theta + \cos^2\theta = 1$.

iii) $\tan^2\theta + 1 = \sec^2\theta$.

iv) $1 + \cot^2\theta = \csc^2\theta$.

v) $\sin(-\theta) = -\sin\theta$.

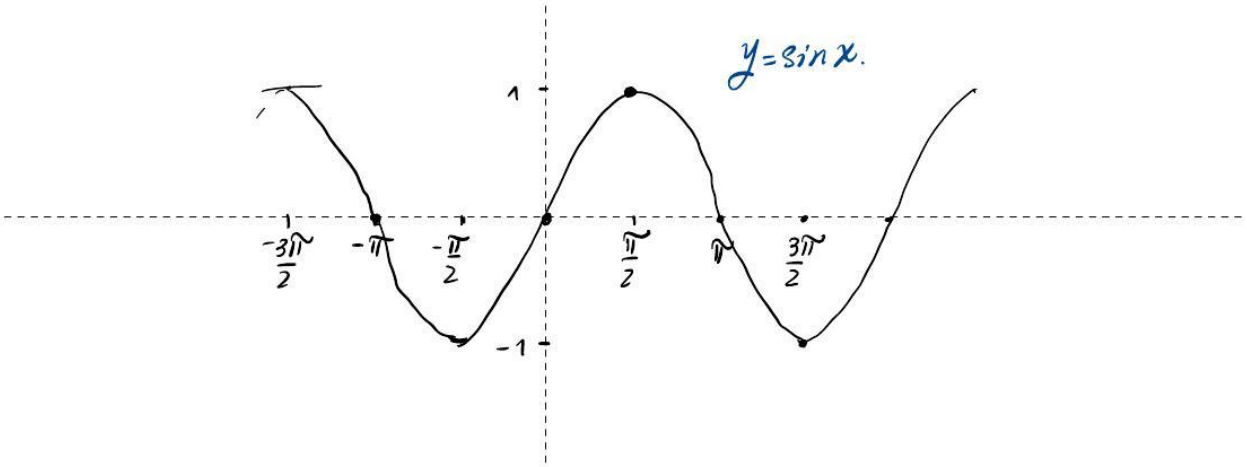
vi) $\cos(-\theta) = \cos(\theta)$

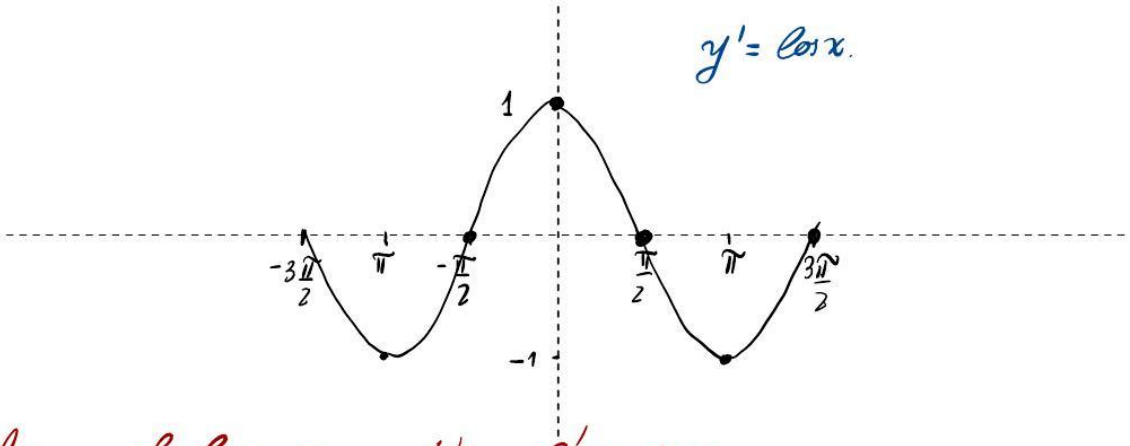
vii).
- $\sin(x+y) = \sin x \cos y + \sin y \cos x$.
- $\sin(x-y) = \sin x \cos y - \sin y \cos x$.

- $\cos(x+y) = \cos x \cos y - \sin x \sin y$.
- $\cos(x-y) = \cos x \cos y + \sin x \sin y$.

viii). $\sin 2x = 2\sin x \cos x$.

2.4. Derivatives of trigonometric functions.





Claim. if $f(x) = \sin x$ then $f'(x) = \cos x$.

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h \to 0} \frac{\sin x \cos h + \sin h \cos x - \sin x}{h}$$

$$= \lim_{h \to 0} \left[\frac{\sin x \cos h - \sin x}{h} + \frac{\sin h \cos x}{h} \right]$$

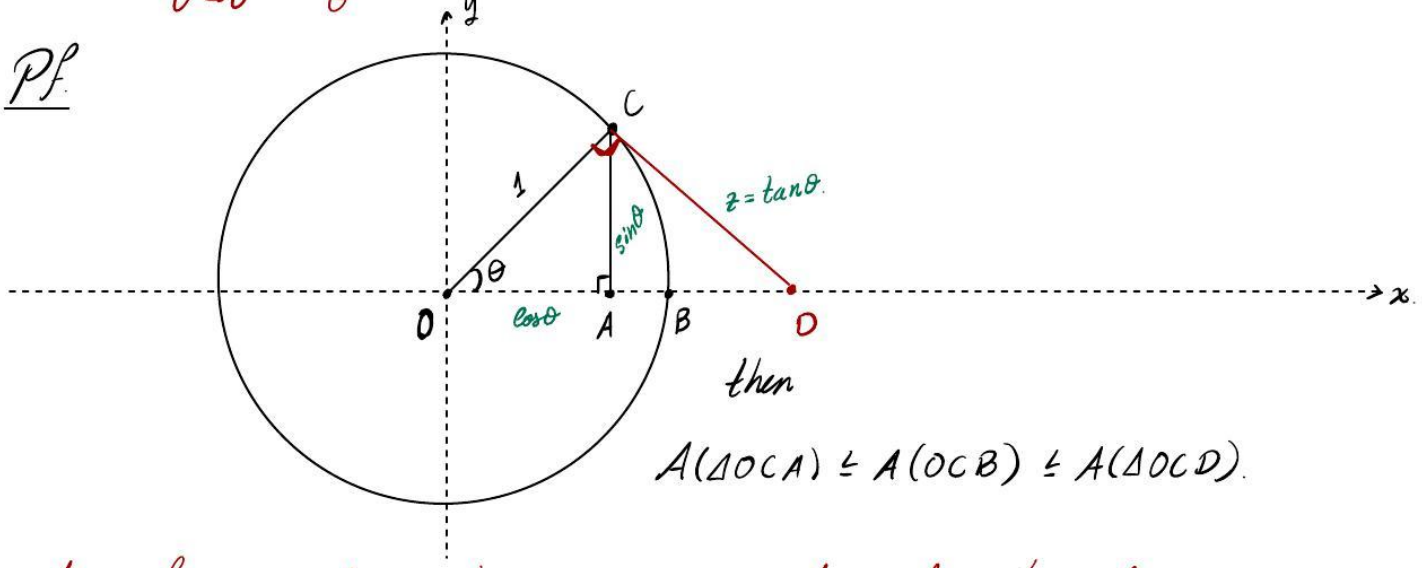
$$= \lim_{h \to 0} \left[\sin x \frac{(\cos h - 1)}{h} + \cos x \cdot \frac{\sin h}{h} \right]$$

$$= \lim_{h\to 0} \sin x \frac{\cos h - 1}{h} + \lim_{h\to 0} \cos x \frac{\sin h}{h}$$

$$= \sin x \cdot \underbrace{\lim_{h\to 0} \frac{\cos h - 1}{h}}_{0} + \cos x \underbrace{\lim_{h\to 0} \frac{\sin h}{h}}_{1}.$$

Thm. $\lim_{\theta\to 0} \frac{\sin\theta}{\theta} = 1.$

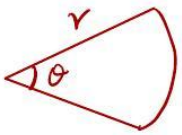
Pf.



then

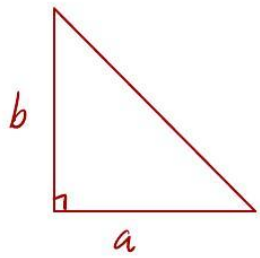
$$A(\Delta OCA) \le A(OCB) \le A(\Delta OCD).$$

Area of a circular sector.



$$A = \frac{\theta \cdot r^2}{2}.$$

Area of a triangle.



$$A = \frac{a \times b}{2}.$$

$$A(\Delta OCA) = \frac{\cos\theta \sin\theta}{2}; \quad A(OCB) = \frac{\theta \cdot 1^2}{2} = \frac{\theta}{2}; \quad \text{Area}(\Delta OCD) = \frac{\tan\theta}{2}.$$

then $$\frac{\cos\theta \sin\theta}{2} \le \frac{\theta}{2} \le \frac{\tan\theta}{2}.$$

$$\cos\theta \sin\theta \le \theta \le \frac{\sin\theta}{\cos\theta} \quad \text{by} \quad a \le b \Rightarrow \frac{1}{b} \le \frac{1}{a}$$

$$\frac{\cos\theta}{\sin\theta} \le \frac{1}{\theta} \le \frac{1}{\cos\theta \sin\theta}.$$

$$\cos\theta \le \frac{\sin\theta}{\theta} \le \frac{1}{\cos\theta}.$$

and $\lim_{\theta \to 0} \cos\theta = 1$, $\lim_{\theta \to 0} \frac{1}{\cos\theta} = 1$. By S-T we get

$$\lim_{\theta \to 0} \frac{\sin\theta}{\theta} = 1.$$

Thm. $\lim_{\theta \to 0} \frac{\cos\theta - 1}{\theta} = 0$

This implies that

$$f'(x) = \sin x \lim_{h \to 0} \frac{\cos h - 1}{h} + \cos x \lim_{h \to 0} \frac{\sin h}{h} = \cos x$$

Example. Differentiate $y = x^2 \sin x$.

Sln. $y = f(x) \cdot g(x)$ where $f(x) = x^2$, $g(x) = \sin x$

$f'(x) = 2x$ $\quad g'(x) = \cos x$

Using the product rule

$$y' = 2x \sin x + x^2 \cos x.$$

Thm. $\frac{d}{dx}(\cos x) = -\sin x.$

Example. Compute $\frac{d}{dx}(\tan x)$.

Sln.

$$\frac{d}{dx}(\tan x) = \frac{d}{dx}\left(\frac{\sin x}{\cos x}\right) = \frac{\frac{d}{dx}(\sin x) \cdot \cos x - \sin x \cdot \frac{d}{dx}(\cos x)}{\cos^2 x}$$

$$= \frac{\cos x \cdot \cos x - \sin x (-\sin x)}{\cos^2 x}$$

$$= \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = \frac{1}{\cos^2 x} = \sec^2 x.$$

Table of derivatives.

$\frac{d}{dx}(\sin x) = \cos x$; $\frac{d}{dx}(\tan x) = \sec^2 x$; $\frac{d}{dx}(\sec x) = \sec x \cdot \tan x$

$\frac{d}{dx}(\cos x) = -\sin x$; $\frac{d}{dx}(\csc x) = -\csc x \cot x$; $\frac{d}{dx}(\cot) = -\csc^2 x$.

Example. if $f(x)=\dfrac{\sec x}{1+\tan x}$ compute f'. For what values of x does the graph of f have horizontal tangent.

Sln. $$f'(x)=\frac{\frac{d}{dx}(\sec x)\cdot(1+\tan x)-\sec x\cdot\frac{d}{dx}(1+\tan x)}{(1+\tan x)^2}.$$

$$=\frac{\sec x\cdot\tan x\,(1+\tan x)-\sec x(\sec^2 x)}{(1+\tan x)^2}$$

$$=\frac{\sec x\,(\tan^2 x+\tan x-\sec^2 x)}{(1+\tan x)^2}.$$

$\tan^2 x+1=\sec^2 x$

$\tan^2 x-\sec^2 x=-1$

$$=\frac{\sec x\,(\tan x-1)}{(1+\tan x)^2}.$$

Notice that $\sec x\neq 0$ then $f'(x)=0$ when $\tan x=1$ or,

$x=\frac{\pi}{4}+n\pi$ where $n\in\mathbb{Z}$.

$\lim\limits_{\theta\to 0}\dfrac{\sin\theta}{\theta}=1$ and $\lim\limits_{\theta\to 0}\dfrac{\cos\theta-1}{\theta}=0.$

Example. find $\lim\limits_{x\to 0}\dfrac{\sin 7x}{4x}$.

Sln. $$\lim_{x\to 0}\frac{\sin 7x}{4x}=\lim_{x\to 0}\frac{7\sin 7x}{4(7x)}=\frac{7}{4}\lim_{x\to 0}\frac{\sin 7x}{7x}=\frac{7}{4}.$$

Example. Calculate $\lim\limits_{x\to 0}x\cdot\cot x$.

Sln. $$\lim_{x\to 0}x\cot x=\lim_{x\to 0}x\frac{\cos x}{\sin x}=\lim_{x\to 0}\frac{\cos x}{\frac{\sin x}{x}}=\frac{\lim\limits_{x\to 0}\cos x}{\lim\limits_{x\to 0}\frac{\sin x}{x}}=\frac{1}{1}=1.$$

Example. Calculate $\lim\limits_{\theta\to 0}\dfrac{\cos\theta-1}{\sin\theta}$

Sln. $$\lim_{\theta\to 0}\frac{\cos\theta-1}{\sin\theta}=\lim_{\theta\to 0}\frac{\frac{\cos\theta-1}{\theta}}{\frac{\sin\theta}{\theta}}=\frac{\lim\limits_{\theta\to 0}\frac{\cos\theta-1}{\theta}}{\lim\limits_{\theta\to 0}\frac{\sin\theta}{\theta}}=\frac{0}{1}=0.$$

2.5 The Chain Rule.

Suppose that you are asked to compute f' where

$$f(x) = \sqrt{x^2+1}$$

Notice that $g(x) = \sqrt{x}$ the outer function and $h(x) = x^2+1$ is the inner function.

and $f(x) = (g \circ h)(x)$.

The Chain Rule: If g is differentiable at x and f is differentiable at $g(x)$ then $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and

$$F'(x) = f'(g(x)) \cdot g'(x).$$

Example. Notice that $F(x) = \sqrt{x^2+1}$ if $f(x) = \sqrt{x}$, $g(x) = x^2+1$ and $F(x) = f(g(x))$. then by the Chain Rule.

$$F'(x) = f'(g(x)) \cdot g'(x)$$

$f'(x) = \frac{1}{2\sqrt{x}}$; $g(x) = 2x$.

$$F'(x) = \frac{1}{2\sqrt{x^2+1}} \cdot (2x) = \frac{x}{\sqrt{x^2+1}}.$$

Rmk. $\frac{d}{dx} f(g(x)) = f'(g(x)) \cdot g'(x)$

- f: outer function
- $g(x)$: evaluated at inner function
- f': derivative of outer function
- $g(x)$: evaluated at inner function
- $g'(x)$: derivative of inner function.

Example. Compute y' if

a) $y = \sin(x^2)$

b) $y = \sin^2(x)$

Sln. a) $y = f(g(x))$ where $f(x) = \sin x$, $g(x) = x^2$.

$f'(x) = \cos x$ $g'(x) = 2x$

$$y' = \cos(x^2)\, 2x = 2x \cos(x^2).$$

b) $y = f(g(x))$ where $f(x) = x^2$ $g(x) = \sin x$.

$f'(x) = 2x$ $g'(x) = \cos x$

$y' = 2 \sin x \cdot \cos x$.

Thm. The power rule combined with the Chain Rule. if n is any real number and $u = g(x)$ then

$$\frac{d}{dx}(u^n) = \frac{d}{dx}(g^n(x)) = n g^{n-1}(x) \cdot g'(x).$$

Example. Compute y' where $y = (x^3 - 1)^{100}$

Sln. $y = f(g(x))$ $f = x^{100}$ $g(x) = x^3 - 1$ $f'(x) = 100\, x^{99}$

$g'(x) = 3x^2$.

$y' = 100(x^3 - 1)^{99} \cdot 3x^2$.

Example. Find $f'(x)$ if $f(x) = \dfrac{1}{\sqrt[3]{x^2 + x + 1}}$

Sln. $f(x) = g(h(x))$ $g(x) = x^{-1/3}$; $h(x) = x^2 + x + 1$.

$g'(x) = -\frac{1}{3} x^{-1/3 - 1} = -\frac{1}{3} x^{-4/3}$ $h'(x) = 2x + 1$.

$$f'(x) = -\frac{1}{3}(x^2 + x + 1)^{-4/3}(2x + 1).$$

Example.
- Find g' if $g(t) = \left(\frac{t-2}{2t+1}\right)^9$
- Compute y' if $y = (2x+1)^5 (x^3 - x + 1)^4$

} for you!

Example. If $F(x) = \sin(\cos(\tan x))$ compute F'.

Sln. $F(x) = f(g(x))$ where $f(x) = \sin x$ $g(x) = \cos(\tan x)$

$g(x) = h(s(x))$ where $h(x) = \cos(x)$ $s(x) = \tan x$.

By the chain Rule.

$$f'(x) = f'(g(x)) \cdot g'(x) \qquad \text{but} \quad g'(x) = h'(s(x))\, s'(x).$$

$$f'(x) = \cos x \; ; \qquad h'(x) = -\sin x \qquad s'(x) = \sec^2 x.$$

$$g'(x) = \cos(\cos(\tan x)) \cdot (-\sin(\tan x))\sec^2 x$$

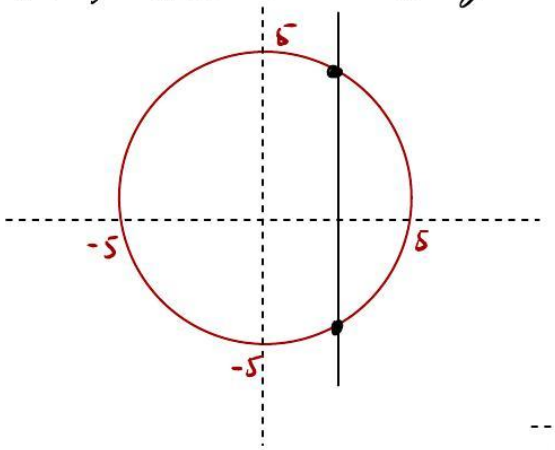
$$= -\cos(\cos(\tan x))\sin(\tan x)\sec^2 x.$$

Problem. Compute y' if $y = \sqrt{\sec(x^3)}$. For you!

We are study functions like

$$y = \sqrt{x^3+1} \qquad y = x\sin x.$$

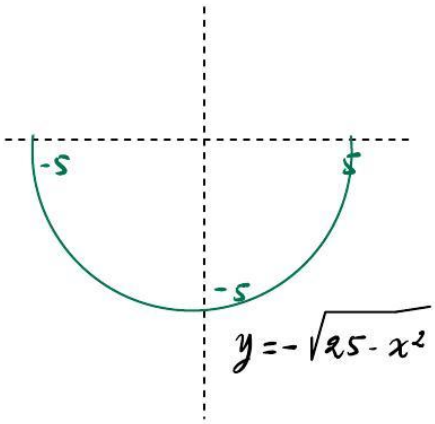
Now, consider $x^2+y^2 = 25$. then.



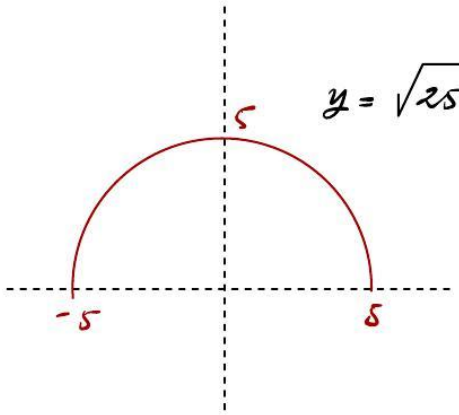
$$x^2+y^2 = 25$$

$$y^2 = 25 - x^2$$

$$y = \pm\sqrt{25-x^2}.$$

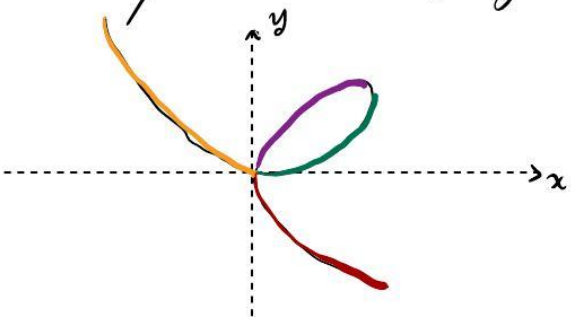


$$y = -\sqrt{25-x^2}$$



$$y = \sqrt{25-x^2}$$

For example let $x^3+y^3 = 6xy$.



Implicit Differentiation. We don't need to solve an equation for y in terms of x in order to find the derivative of y. Instead we can use implicit differentiation. This consist of differentiating both sides of the equation with respect to x and then solving the resulting equation for $\frac{dy}{dx}$.

Example. if $x^2+y^2=25$ find $\frac{dy}{dx}$ and find the equation of the tangent to te circle at $(3,4)$.

Sln. **step 1.** Take $\frac{d}{dx}$ in both sides.

$$\frac{d}{dx}\left(x^2+y^2\right) = \frac{d}{dx}(25)$$

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) = 0.$$

step 2. if you need compute $\frac{d}{dx}(G(y))$ use the chain rule.

$$2x + \frac{d}{dx}(y^2) = 0$$

Let $G(y) = y^2$ $\quad G(y) = H(F(y))$ $\quad H(x) = x^2$ $\quad F(y) = y$.

$$G'(y) = 2(y)\cdot y'$$

$$2x + 2yy' = 0$$

Step 3. solve for y'

$$2x + 2y\,y' = 0$$

$$2y\,y' = -2x$$

$$y' = \frac{-2x}{2y} = -\frac{x}{y}.$$

$P(3,4)$ $\quad x_1\ y_1$ $\quad$ my slope. $\quad m = y'\Big|_{(3,4)} = -\frac{3}{4}.$

Using slope-point

$$y - y_1 = m(x - x_1)$$

$$y - 4 = -\frac{3}{4}(x - 3)$$

$$y = -\frac{3}{4}x + \frac{25}{4}.$$